

$$1. \quad k = -\frac{1}{2\sqrt{EG}} \left(\frac{\partial}{\partial u} \left(\frac{G_u}{\sqrt{EG}} \right) + \frac{\partial}{\partial v} \left(\frac{E_v}{\sqrt{EG}} \right) \right)$$

$\therefore (u, v)$ 正交坐标

$$\therefore F=0$$

$$\Gamma_{11}^1 = \frac{GE_u - 2FF_u + FE_v}{2(EG - F^2)} = \frac{E_u}{2E}$$

$$\Gamma_{11}^2 = \frac{2EF_u - EE_v - FE_u}{2(EG - F^2)} = \frac{-E_v}{2G}$$

$$(\Gamma_{11}^2)_v = -\frac{1}{2} \frac{E_{vv}G - E_vG_v}{G^2}$$

$$\Gamma_{1v}^1 = \frac{GE_v - FG_u}{2(EG - F^2)} = \frac{E_v}{2E}$$

$$\Gamma_{1v}^2 = \frac{EG_u - FE_v}{2(EG - F^2)} = \frac{G_u}{2G}$$

$$(\Gamma_{1v}^2)_u = \frac{1}{2} \frac{G_{uu}G - G_u^2}{G^2}$$

$$\Gamma_{22}^1 = \frac{2GF_v - GG_u - FG_v}{2(EG - F^2)} = -\frac{G_u}{2E}$$

$$\Gamma_{2v}^2 = \frac{EG_v - 2FF_v + FG_u}{2(EG - F^2)} = \frac{G_v}{2G}$$

Gauss 方程:

$$EF = (\Gamma_{11}^2)_v - (\Gamma_{1v}^2)_u + \Gamma_{11}^1 \Gamma_{1v}^2 + \Gamma_{11}^2 \Gamma_{22}^2 - \Gamma_{1v}^1 \Gamma_{11}^2 - \Gamma_{1v}^2 \Gamma_{12}^2$$

$$= -\frac{E_{vv}G - E_vG_v}{2G^2} - \frac{G_{uu}G - G_u^2}{2G^2} + \frac{E_uG_u}{4EG} - \frac{E_vG_v}{4G^2} - \frac{G_u^2}{4G^2}$$

$$k = -\frac{1}{2\sqrt{EG}} \left(\frac{E_{vv}G - E_vG_v}{\sqrt{E}\sqrt{G}G} + \frac{G_{uu}G - G_u^2}{\sqrt{E}\sqrt{G}G} - \frac{1}{2} \frac{E_uG_u}{\sqrt{E}\sqrt{G}E} + \frac{1}{2} \frac{E_vG_v}{\sqrt{E}\sqrt{G}G} + \frac{1}{2} \frac{G_u^2}{\sqrt{E}\sqrt{G}G} \right)$$

$$= -\frac{1}{2\sqrt{EG}} \left(\frac{E_{vv}}{\sqrt{EG}} - \frac{E_vG_v}{2G\sqrt{EG}} + \frac{G_{uu}}{\sqrt{EG}} - \frac{G_uG_u}{2G\sqrt{EG}} - \frac{E_uG_u}{2E\sqrt{EG}} + \frac{E_vG_v}{2G\sqrt{EG}} \right)$$

$$\frac{\partial}{\partial u} \left(\frac{G_u}{\sqrt{EG}} \right) = \frac{G_{uu}\sqrt{EG} - G_u \frac{E_uG + E G_u}{2\sqrt{EG}}}{EG} = \frac{2G_{uu}EG - E_uG_uG - E G_u^2}{2EG\sqrt{EG}}$$

$$= \frac{G_{uu}}{\sqrt{EG}} - \frac{E_uG_u}{2E\sqrt{EG}} - \frac{G_u^2}{2G\sqrt{EG}}$$

$$\frac{\partial}{\partial v} \left(\frac{E_v}{\sqrt{EG}} \right) = \frac{E_{vv} \sqrt{EG} - E_v \frac{E_v G + E G_v}{2 \sqrt{EG}}}{EG} = \frac{2 E_{vv} EG - E_v^2 G - E E_v G_v}{2 EG \sqrt{EG}}$$

$$= \frac{E_{vv}}{\sqrt{EG}} - \frac{E_v^2}{2E \sqrt{EG}} - \frac{E_v G_v}{2G \sqrt{EG}}$$

$$\therefore K = - \frac{1}{2 \sqrt{EG}} \left(\frac{\partial}{\partial u} \left(\frac{G_u}{\sqrt{EG}} \right) + \frac{\partial}{\partial v} \left(\frac{E_v}{\sqrt{EG}} \right) \right)$$

2. $E=1$, $F=0$. 在测地极坐标 (r, θ) 下, $K = - \frac{(\sqrt{G})_{rr}}{\sqrt{G}}$.

$$E = r_r \cdot r_r = 1, \quad F = r_r \cdot r_\theta = 0, \quad G = r_\theta \cdot r_\theta$$

$$\Gamma_{11}^1 = \frac{G E_r - 2 F F_r + F E_\theta}{2(EG - F^2)} = 0$$

$$\Gamma_{11}^2 = \frac{2 E F_r - E E_\theta - F E_r}{2(EG - F^2)} = 0$$

$$\Gamma_{12}^1 = \frac{G E_\theta - F G_r}{2(EG - F^2)} = 0$$

$$\Gamma_{12}^2 = \frac{E G_r - F E_\theta}{2(EG - F^2)} = \frac{G_r}{2G}$$

$$\Gamma_{22}^1 = \frac{2 G F_\theta - G G_r - F G_\theta}{2(EG - F^2)} = \frac{-G_r}{2}$$

$$\Gamma_{22}^2 = \frac{E G_\theta - 2 F F_\theta + F G_r}{2(EG - F^2)} = \frac{G_\theta}{2G}$$

$$K = (\Gamma_{11}^2)_\theta - (\Gamma_{12}^2)_r + \Gamma_{11}^1 \Gamma_{12}^2 + \Gamma_{11}^2 \Gamma_{22}^2 - \Gamma_{12}^1 \Gamma_{11}^2 - \Gamma_{12}^2 \Gamma_{12}^2$$

$$= 0 - \left(\frac{G_r}{2G} \right)_r - \left(\frac{G_r}{2G} \right)^2$$

$$= - \frac{2G G_{rr} - G_r G_r}{2G^2}$$

$$(\sqrt{G})_{rr} = \frac{G_r}{2 \sqrt{G}}$$

$$(\sqrt{G})_{PP} = \frac{1}{2} \frac{G_{PP}\sqrt{G} - G_P \frac{G_P}{2\sqrt{G}}}{G}$$

$$= \frac{2G G_{PP} - G_P^2}{4G \sqrt{G}}$$

$$\therefore K = - \frac{(\sqrt{G})_{PP}}{\sqrt{G}}$$

单位球面 $S^2 = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1\}$

(a) $P = (0, 0, 1) \quad \exp_P : T_P S^2 \rightarrow S^2$

$$T_P S^2 = \{(u, v, 0) \mid u, v \in \mathbb{R}\}$$

$$\exp_P(u, v) = \left(\frac{u}{\sqrt{u^2+v^2}} \sin \sqrt{u^2+v^2}, \frac{v}{\sqrt{u^2+v^2}} \sin \sqrt{u^2+v^2}, \cos \sqrt{u^2+v^2} \right)$$

(b) 在 $T_P S^2$ 中: $v = (u, v) = r(\cos \theta, \sin \theta)$

$$\exp_P(r \cos \theta, r \sin \theta) = (\cos \theta \sin r, \sin \theta \sin r, \cos r)$$

(c) $\exp_P(u, v) = \left(\frac{\sin \|v\|}{\|v\|} u, \frac{\sin \|v\|}{\|v\|} v, \cos \|v\| \right)$

① $0 \leq \|v\| < \pi$ ($\exp_P(u, v)$ 为双射)

② $\|v\| = \pi \quad \exp_P(u, v) = (0, 0, 1) \quad \text{不是双射}$

③ $\|v\| > \pi \quad \wedge \exists \|v\| = 2\pi - s \quad s \in [0, \pi)$

$$\exp_P(u, v) = \left(\frac{-\sin s}{2\pi - s} u, \frac{-\sin s}{2\pi - s} v, \cos s \right)$$

与 $\|v\| \in [0, \pi)$ 某点会重合. 不是双射

$$\therefore 0 \leq \|v\| < \pi$$

1. Gauss 曲率 $K = 10000$ 的常曲率曲面局部同构于 $R = ?$ 的球面

$$r(\theta, \varphi) = (R \cos \theta \cos \varphi, R \sin \theta \cos \varphi, R \sin \varphi) \quad \theta \in [0, 2\pi), \varphi \in [0, \pi]$$

$$r_\theta = (-R \sin \theta \cos \varphi, R \cos \theta \cos \varphi, 0) \quad r_{\theta\theta} = (-R \cos \theta \cos \varphi, -R \sin \theta \cos \varphi, 0)$$

$$r_\varphi = (-R \cos \theta \sin \varphi, -R \sin \theta \sin \varphi, R \cos \varphi) \quad r_{\varphi\varphi} = (R \sin \theta \sin \varphi, -R \cos \theta \sin \varphi, 0)$$

$$E = r_\theta \cdot r_\theta = R^2 \cos^2 \varphi$$

$$r_{\varphi\varphi} = (-R \cos \theta \cos \varphi, -R \sin \theta \cos \varphi, -R \sin \theta \sin \varphi)$$

$$F = r_\theta \cdot r_\varphi = 0$$

$$G = r_\varphi \cdot r_\varphi = R^2$$

$$EG - F^2 = R^4 \cos^2 \varphi.$$

$$\vec{N} = \frac{r_\theta \times r_\varphi}{\|r_\theta \times r_\varphi\|} = (\cos \theta \cos \varphi, \sin \theta \cos \varphi, \sin \varphi)$$

$$r_\theta \times r_\varphi = \begin{vmatrix} i & j & k \\ -R \sin \theta \cos \varphi & R \cos \theta \cos \varphi & 0 \\ -R \cos \theta \sin \varphi & -R \sin \theta \sin \varphi & R \sin \varphi \end{vmatrix}$$

$$= i (R^2 \cos \theta \cos^2 \varphi) - j (-R^2 \sin \theta \cos^2 \varphi) + k (R^2 \sin^2 \theta \sin \varphi \cos \varphi + R^2 \cos^2 \theta \sin \varphi \cos \varphi)$$

$$= (R^2 \cos \theta \cos^2 \varphi, R^2 \sin \theta \cos^2 \varphi, R^2 \sin \varphi \cos \varphi)$$

$$= R^2 \cos \varphi (\cos \theta \cos \varphi, \sin \theta \cos \varphi, \sin \varphi)$$

$$\|r_\theta \times r_\varphi\| = (R^4 \cos^2 \theta \cos^4 \varphi + R^4 \sin^2 \theta \cos^4 \varphi + R^4 \sin^2 \varphi \cos^2 \varphi)^{\frac{1}{2}}$$

$$= (R^4 \cos^2 \varphi \cos^2 \varphi + R^4 \sin^2 \varphi \cos^2 \varphi)^{\frac{1}{2}}$$

$$= (R^4 \cos^2 \varphi)^{\frac{1}{2}}$$

$$= R^2 \cos \varphi$$

$$L = \vec{N} \cdot r_{\theta\theta} = (\cos \theta \cos \varphi, \sin \theta \cos \varphi, \sin \varphi) \cdot (-R \cos \theta \cos \varphi, -R \sin \theta \cos \varphi, 0)$$

$$= -R \cos^2 \varphi$$

$$M = \vec{N} \cdot r_{\varphi\varphi} = (\cos \theta \cos \varphi, \sin \theta \cos \varphi, \sin \varphi) \cdot (R \sin \theta \sin \varphi, -R \cos \theta \sin \varphi, 0) = 0$$

$$N = \vec{N} \cdot r_{\varphi\varphi} = (\cos \theta \cos \varphi, \sin \theta \cos \varphi, \sin \varphi) \cdot (-R \cos \theta \cos \varphi, -R \sin \theta \cos \varphi, -R \sin \varphi)$$

$$= -R$$

$$LN-M^2 = R^2 \omega^2 \rho$$

$$K = \frac{LN-M^2}{EG-F^2} = \frac{1}{R^2} = 1000.$$

$$R = \frac{1}{100}$$

2. 证明 Poincaré 圆盘的 Gauss 曲率为 -1.

$$E = \frac{4}{(1-u^2-v^2)^2} = 4, \quad F = 0$$

$$K = -\frac{1}{2\sqrt{EG}} \left(\frac{\partial}{\partial u} \left(\frac{G_u}{\sqrt{EG}} \right) + \frac{\partial}{\partial v} \left(\frac{F_v}{\sqrt{EG}} \right) \right)$$

$$E_v = \left(\frac{4}{(1-u^2-v^2)^2} \right)_v = 4 \frac{-2(1-u^2-v^2) \cdot (-2v)}{(1-u^2-v^2)^4} = \frac{16v}{(1-u^2-v^2)^3}$$

$$G_u = \left(\frac{4}{(1-u^2-v^2)^2} \right)_u = \frac{16u}{(1-u^2-v^2)^3}$$

$$\frac{G_u}{\sqrt{EG}} = \frac{\frac{16u}{(1-u^2-v^2)^3}}{\frac{4}{(1-u^2-v^2)^2}} = \frac{4u}{1-u^2-v^2}$$

$$\left(\frac{G_u}{\sqrt{EG}} \right)_u = \frac{4(1-u^2-v^2) - 4u(-2u)}{(1-u^2-v^2)^2} = \frac{4 - 4u^2 - 4v^2 + 8u^2}{(1-u^2-v^2)^2} = \frac{4(1+u^2-v^2)}{(1-u^2-v^2)^2}$$

$$\left(\frac{F_v}{\sqrt{EG}} \right)_v = \frac{4(1-u^2+v^2)}{(1-u^2-v^2)^2}$$

$$K = -1$$

3. 在 Poincaré 上半平面模型中取 $y=1$. 证明它是测地线

$$ds^2 = \frac{dx^2 + dy^2}{y^2}, \quad E = G = \frac{1}{y^2} = 1, \quad F = 0$$

$$r(t) = (u(t), v(t)) = (t, 1)$$

$$\left\{ \begin{aligned} \frac{d}{dt} (E\ddot{u} + F\ddot{v}) &= \frac{1}{2} (E_u \dot{u}^2 + 2F_u \dot{u}\dot{v} + G_u \dot{v}^2) \quad (*) \end{aligned} \right.$$

$$\left\{ \begin{aligned} \frac{d}{dt} (F\dot{u} + G\dot{v}) &= \frac{1}{2} (E_v \dot{u}^2 + 2F_v \dot{u}\dot{v} + G_v \dot{v}^2) \quad (**) \end{aligned} \right.$$

$$(*) \quad \ddot{r} = \frac{d}{dt} (1+0) = 0 \quad \text{在}$$

$$(**) \quad \ddot{r} = \frac{d}{dt} (0+0) = 0$$

$$r = \frac{1}{2} (-2 \times 1 + 0 + (-2) \times 0) = -1 \neq 0$$

$\therefore$ 不是测地线

$$\begin{aligned} 1. \quad dw &= d\left(\frac{z-i}{z+i}\right) \\ &= \frac{z+i - (z-i)}{(z+i)^2} dz \\ &= \frac{2i}{(z+i)^2} dz \\ &= \frac{2i}{(x+iy+1)^2} d(x+iy) \end{aligned}$$

$$2. \quad p(x_0, y_0, z_0) \quad x_0^2 + y_0^2 - z_0^2 = -1$$

$$\langle p, v \rangle = x_0 x + y_0 y - z_0 z = 0$$

$$\text{即 } x_0 x + y_0 y = z_0 z$$

$$(x_0 x + y_0 y)^2 = z_0^2 z^2$$

$$\text{左: } (x_0 x + y_0 y)^2 \leq (x_0^2 + y_0^2)(x^2 + y^2)$$

$$\text{右: } z_0^2 z^2 = (x_0^2 + y_0^2 + 1) z^2$$

$$(x_0^2 + y_0^2)(x^2 + y^2) \geq (x_0^2 + y_0^2 + 1) z^2$$

$$(x_0^2 + y_0^2)(x^2 + y^2 - z^2) \geq z^2 > 0$$

$$\therefore x^2 + y^2 - z^2 > 0$$

$$\langle 2 \rangle \quad e_1 = (1, 0, 0) \quad e_2 = (0, 0, 1)$$

$$\langle e_1, e_1 \rangle = 1 \quad \langle e_1, e_2 \rangle = 0 \quad \langle e_2, e_2 \rangle = -1$$

$$K = \frac{\langle R(e_1, e_2)e_1, e_1 \rangle}{\langle e_1, e_1 \rangle \langle e_2, e_2 \rangle - \langle e_1, e_2 \rangle^2}$$

$$= \frac{\langle N, N \rangle}{-1} \frac{(B(e_1, e_2) B(e_2, e_1) - B(e_1, e_1) B(e_2, e_2))}{-1}$$

$$= \frac{-1 \cdot (-1)}{-1}$$

$$= -1$$